Good Bug, Bad Bug: Breaking through Microbial Stereotypes

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Our expanding knowledge of microbial mechanisms is challenging the notion of "good" versus "bad" microbes and encouraging a better understanding of their roles in various contexts before their widespread therapeutic and clinical application. The intestinal microbe *Akkermansia muciniphila*, a promising probiotic with an emerging cautionary tale, best highlights this challenge.

The recent emergence of the "hygiene hypothesis" demonstrates a shift in mindset from believing that cleaner is always better to recognizing the vital role microbes play in human health and development. Today, as rapid advances continue to be made in our understanding of the human microbiome, parallel efforts are taking place to design microbe-based interventions to combat disease and improve health. These efforts are typically focused on microbes broadly considered to be "beneficial," such as certain strains of *Bifidobacterium*, *Lactobacillus*, and *Akkermansia*. Meanwhile, there exists a contrasting assortment of microbes generally thought of as directly "harmful" or that are associated with dysbiosis, but which should not be entirely neglected in the field of microbial therapeutics. In both cases, much of the evidence by which microbes are (often subconsciously) classified as "good" or "bad' is based on observations in highly specific disease settings, and it is worth noting that knowledge of how a microbe interacts with its host in one context may not always broadly apply to all aspects of health.

Microbe-Based Therapies: A Double-Edged Sword

On a microbial-population level, this concept has been familiar for some time. It is now nearly indisputable that fecal microbiota transplantation (FMT) is a highly effective therapy for *Clostridium difficile* infection and represents a life-changing intervention for many individuals afflicted with chronic recurrent *C. difficile* infections (Kelly et al., 2015). This success is driving efforts to apply FMT (and collections of isolated microbes) to other conditions, such as irritable bowel syndrome, and even to systemic metabolic disorders. However, it has also been recognized that long-term studies are required to ensure that FMT does not transfer undetected and potentially pathogenic viruses, bacteria, or parasites and other biological materials from donors to recipients. Likewise, the increasing evidence of gut microbiome involvement in various conditions, including colon cancer, depression, and obesity, suggests that care should be taken to ensure that FMT does not induce a shift in the recipient host microbiota that puts them at increased risk for these conditions (Kelly et al., 2015). Concurrent efforts to optimize donor selection, improve screening of transferred microbes (especially protists and viruses), conduct long-term follow-up studies of recipients, and even to develop entirely *in vitro* cultures for transfer will ensure that FMT has an exciting and safe future for a variety of indications.

This holistic approach to the study of microbe-based interventions should also be applied to individual microbes, especially when therapies are tailored to be efficacious for a specific disease or phenotype. *Clostridium* clusters IV and XIVa can induce the production of anti-inflammatory molecules, such as interleukin-10, and enhance the abundance of regulatory T cells (Tregs) in their host; when introduced into mouse models, they are consequently able to attenuate symptoms of colitis and certain autoimmune and allergic conditions (Atarashi et al., 2013). However, the broad and long-term consequences of such a shift in the host immune system require close inspection. Although beneficial in colitis and autoimmune disorders, Tregs may limit the immune response against tumors, and elevated Treg levels are associated with poor prognosis in certain cancers (Adeegbe and Nishikawa, 2013). Likewise, the production of secondary bile acids by these *Clostridium* clusters may be the mechanism by which their effect on immune regulation is modulated, yet secondary bile acids have been associated with colon cancer (Bernstein et al., 2011). As animal models of colitis and cancer rarely overlap, and the experimental time frame of microbiota-modulating animal studies is generally short, such unintended consequences may not be easily recognized.

Gloomy Microbes, Golden Opportunities

The reverse phenomenon also exists, whereby microorganisms colloquially considered harmful or pathogenic have important beneficial functions. A classic example is the bacterium *Escherichia coli*, which, depending on the strain, can be an extremely virulent pathogen that causes acute hemorrhagic diarrhea and haemolytic uremic syndrome but also a normal constituent of the gut microbiota, a probiotic, or a model research organism and industrial workhorse used to produce important commercial therapeutics such as recombinant insulin. Incidentally, although this is broadly appreciated in the case of *E. coli*, it is typically neglected in microbiome studies that primarily sequence short variable regions of the 16S rRNA gene to identify bacterial taxa.

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Depending on the microorganism and the analysis tools used, 16S sequencing usually enables taxonomic identification down to at least the family level but is very rarely able to distinguish between different strains of the same species or related species (for example, *Salmonella*, *E. coli*, and *Shigella* would all fall under the same 16S sequence). The example made salient by *E. coli* applies to all microbes; identical 16S sequences can represent entirely different bacteria with drastically different phenotypes, especially as they relate to human health. Horizontal gene transfer between different microbes densely populating the same habitat, such as the human gut, further complicates this issue.

More nuanced examples of the potential benefits of microorganisms normally associated with harm or dysbiosis also exist in the realm of host-microbe interactions. Often, the relative risk or advantage to harboring certain microbial species depends on external factors. *Helicobacter pylori*, a Gram-negative bacterium that is able to colonize the human stomach, has a well-established role as a major causative agent of peptic ulcers and gastric carcinoma. The ability of this organism to penetrate the mucosal layer of the stomach and produce urea to increase the pH in its immediate surroundings allows it to persist within the highly acidic environment of the stomach. Unfortunately, this can also contribute to tissue damage, resulting in the formation of ulcers. As the damaged tissue heals and cells regenerate, pro-cancerous mutations may be acquired with the risk of the development of deleterious mutations increasing with age as the cycle of damage and regeneration continues. This provides a clear mechanism for the role of *H. pylori* in disease in adults and reveals the reasoning behind using antibiotics for its eradication. However, this microbe has coevolved with humans for tens of thousands of years and colonizes around half of all humans without causing disease in the majority of cases, and multiple studies have found an association between the presence of *H. pylori* in the stomach and a decreased risk of early-onset asthma and other allergic disorders in children (reviewed in Amedei et al., 2010). This points to a beneficial role for this microbe in normal immune development during early life. Studies in animal models have demonstrated a possible mechanism for this that involves promoting a Th1 immune response over the alternative Th2 allergic immune response (Amedei et al., 2010). Investigations into the soundness of this association, as well as the underlying mechanism, are ongoing and controversial; however, these initial findings highlight the complexity of human-microbiota interactions and the importance of taking confounding factors, such as host age, host genetics, and environment, into account.

Such complex relationships are not constrained to bacteria. Helminths, a broad group of organisms including hookworms, roundworms, schistosomes, and tapeworms, are typically considered intestinal parasites. Not unlike *H. pylori*, there is a long co-evolutionary history between host and infecting organism, and most infected individuals are asymptomatic. Nevertheless, the level of worm-associated morbidity can be quite high in low-resource areas and in immunocompromised individuals, and therefore deworming is both advocated and practiced in many settings (Wammes et al., 2014). Type 2 immune responses, associated with Th2 cells, are the body's main tool to combat helminth colonization and proliferation; as such, helminths have evolved mechanisms by which to dampen these responses in their host while enhancing regulatory Treg capacity. This ability of helminths to modulate the immune system has led to the hypothesis that helminth infection may protect against inflammatory diseases via modulation of the host immune system to a more tolerant state. Indeed, clinical trials of live helminth (or helminth ova/larvae) infection are currently underway for allergy, asthma, multiple sclerosis, inflammatory bowel disease, type 1 diabetes, Crohn's disease, celiac disease, colitis, and rheumatoid arthritis. Some of these trials are beginning to show positive results; perhaps the most promising is the emerging evidence that *Trichuris suis* ova infection may help combat Crohn's disease. *Necator americanus* (human hookworm) larvae present another interesting example. A high inoculation rate results in substantial detrimental gastrointestinal symptoms, but a smaller dose may help lower the host immune response to gluten in celiac disease, although these observations remain preliminary (Wammes et al., 2014).

The field of virology has embraced a similar concept, with the recognition of symbiotic viruses that establish chronic, asymptomatic infections, thereby modulating the host immune system and increasing resistance to other infections, diseases, or environmental stressors. An example of this is the greater resistance to bacterial infection and tumor grafts found in mice with chronic γ-herpesvirus infection (Virgin, 2014). A healthy human is estimated to have over ten constant chronic viral infections (Virgin, 2014); since these infections can persist systemically, this provides a wide network of long-lasting interactions between viruses and their hosts. Moreover, this value is expected to be an underestimate, and advancements in technologies for characterizing the virome will likely bring increased appreciation for its role within the healthy human microbiome. Acutely pathogenic viruses also have benefits to offer, with the development of oncolytic viruses as anti-cancer tools showing accelerating promise. Similar to harnessing the naturally immuno-modulatory properties of helminths, here researchers often exploit the naturally pathogenic properties of a virus for therapeutic advantage. The devastating neurological effects of poliovirus have led to a massive global eradication initiative; yet, simultaneously, its natural preference to target the nervous system has made it an obvious choice to develop therapies against glioblastoma, a type of brain cancer (Denniston et al., 2016). In the case of oncolytic viruses, genetic modifications are universally necessary to render the virus unable to cause disease, yet many of the viral properties inherent to their pathogenicity (specific cell targeting, rapid replication, potent host cell toxicity, ability to evade the immune system, and immunomodulatory effects) are quintessential to their therapeutic application.

The Curious Case of *Akkermansia*

Perhaps the most notable example of the "good bug, bad bug" phenomenon is the intestinal bacterium *Akkermansia muciniphila*, a mucosal-dwelling anaerobe and the only known member of its genus. Since its discovery in 2004, it has garnered a great deal of scientific interest due the consistent observation that intestinal *Akkermansia* abundance is reduced in a variety of metabolic disorders, including

obesity, dyslipidemia, and type 2 diabetes, in dozens of animal and human studies (reviewed in Derrien et al., 2017). These observations have stimulated several follow-up mechanistic studies, which have demonstrated that *Akkermansia* supplementation restores epithelial mucosal integrity, reduces weight gain and fat accumulation, improves glucose tolerance, and reduces inflammation and metabolic endotoxemia in animal models of diabetes and obesity (Derrien et al., 2017; Plovier et al., 2017). Spurred by these exciting results, there is widespread interest in designing *Akkermansia*-based probiotic therapies to combat metabolic disorders and improve health in humans. As the only known intestinal representative of an entire phylum (Verrucomicrobia), *Akkermansia muciniphila* contains many genes unique among the human microbiome, and altering its abundance therefore has the potential to be especially impactful on the functional profile of the entire microbiota. Indeed, it has recently been shown that Amuc_1100, an *Akkermansia*-specific outer membrane protein, recapitulates many of the beneficial effects of the live bacterium in diet-induced obesity (Plovier et al., 2017), and efforts are underway to translate this into a human therapeutic.

However, recent findings from the field of neurology provide a cautionary note. Since 2015, six studies examining the gut microbiome in Parkinson's disease (PD) have reported an increase in *Akkermansia* abundance in PD patients (e.g., Heintz-Buschart et al., 2017; Hill-Burns et al., 2017); indeed, elevated *Akkermansia* abundance appears to be the most consistently defining feature of the PD microbiome. Likewise, a 2017 study found elevated *Akkermansia* in individuals with rapid eye movement sleep behavior disorder, which is considered a pre-motor symptom of PD (Heintz-Buschart et al., 2017). Additionally, several emerging studies (e.g., Berer et al., 2017; Cekanaviciute et al., 2017) have reported an increased abundance of *Akkermansia* in multiple sclerosis (MS), one of which enrolled monozygotic twins discordant for MS diagnosis, thereby ruling out genetic confounders. Both studies also demonstrated that fecal microbiota samples from MS-affected individuals caused worse disease progression than those from control individuals when transplanted into a mouse model of MS. Most recently, the first comprehensive study of the gut microbiome in human Alzheimer's disease (AD) also observed increased *Akkermansia* in AD patients compared to controls (Vogt et al., 2017). Potential causal mechanisms underpinning these observations have not yet been explored, although PD, AD, and MS all involve important immune components as well as metabolic alterations. PD is especially closely connected to the gut microbiome, with the majority of patients experiencing chronic constipation and other gastrointestinal comorbidities. Although speculative, it is not unreasonable to hypothesize that the TLR2 activation induced by Amuc_1100 (Plovier et al., 2017) or the modulation of glucose and cholesterol homeostasis (Derrien et al., 2017) induced by *Akkermansia* may have unforeseen deleterious consequences for neurological health in certain individuals. Notably, although now nearly a dozen papers exist, to our knowledge, there were no published studies reporting a correlation between elevated *Akkermansia* abundance and the incidence of any neurological condition prior to 2014, and no mechanistic studies to date have explored this association in detail. As our understanding of the gut-brain axis is in its infancy relative to investigations into the microbiome and metabolic disorders, this is perhaps not surprising, yet it highlights that novel aspects of the human-microbiota relationship will undoubtedly be revealed through continued exploration by the scientific community, which will lead to further modifications in our perception of the role certain microbes play in human health.

Final Thoughts

As seen with helminths and *H. pylori*, efforts to investigate novel and potentially unanticipated benefits of stereotypically harmful microbes should continue. Likewise, while there is justifiable excitement for microbial therapies such as FMT and *Clostridium*- or *Akkermansia*-based probiotics, their broad effects on the host beyond the conditions they are designed to treat require careful holistic investigation, including long-term follow-up studies. Better strain-level identification of bacteria in the human microbiome will also help unravel some of this complexity. It is increasingly clear that whether a microbe is helpful or harmful is highly dependent on a multitude of host and microbial factors, including the dose, strain, duration of colonization, and specific application of the microbe; and the age, genetic background, environmental factors, disease status, and host process in question. Ultimately, these factors should be closely considered in the design and trial of microbial interventions for human therapeutic use. The answer to the "good bug, bad bug" question is not dichotomous, but rather one of context.

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